

NOTES FOR MAT1101

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1. LECTURE 1

1.1. Motivation for Galois Theory. The first part of the course is Galois theory, for which the original motivation was to understand solvability of polynomials using radicals. Galois had the insight that one should aim to understand the symmetries among the roots.

Example 1.1. Consider the polynomial equation $x^3 - 3x + 1 = 0$ which is irreducible over $\mathbb{Q}$. If $\alpha \in \mathbb{C}$ is a root, then the others are given by $1 - \frac{1}{\alpha}$ and $\frac{1}{1-\alpha}$. This situation is not typical for cubic polynomials. One can then form the subfield $\mathbb{Q}(\alpha) \subseteq \mathbb{C}$ generated by the roots. Since $\alpha^3 = 3\alpha - 1$ we find that

$$\mathbb{Q}(\alpha) = \{a + b\alpha + c\alpha^2 : a, b, c \in \mathbb{Q}\}.$$

The symmetries of this field extension over $\mathbb{Q}$ are uniquely defined by the image of α among $\alpha, 1 - \frac{1}{\alpha}$ and $\frac{1}{1-\alpha}$. Hence, the symmetries form the group $\mathbb{Z}/3\mathbb{Z}$.

In general, if $K \subseteq L$ is a field extension, we can define

$$\text{Gal}(L/K) := \{\sigma : L \rightarrow L \text{ field isomorphism} \mid \forall a \in K : \sigma(a) = a\}.$$

One goal of the course will be to show that if $K \subseteq L$ is sufficiently nice (finite and Galois), we get a canonical bijection

$$\{\text{intermediate fields in } K \subseteq L\} \longleftrightarrow \{\text{subgroups of } \text{Gal}(L/K)\}.$$

For future reference, in this course all rings will have a 1.

1.2. Recollection from MAT1100. Let us recall a couple of facts about fields from MAT1100.

Exercise 1.2. Show that any homomorphism of fields is injective.

Exercise 1.3. Recall that if k is a field, then we can form the *prime subfield*, which is defined as the intersection of all subfields of k . Show that the prime subfield is either $\mathbb{Q}$ or $\mathbb{F}_p$ for some prime $p > 0$.

We remember that $k[x]$ is a principal ideal domain and that two ideals (f) and (g) are equal if and only if f and g differ by multiplication by a unit. Furthermore, the prime ideals of $k[x]$ are the zero ideal and the ideals (f) where f is a monic irreducible polynomial. Only the latter are maximal.

We also had the factor theorem, telling us that if $\alpha \in k$ and $f(\alpha) = 0$ for some polynomial $f \in k[x]$, then $(x - \alpha)$ divides $f(x)$. As a corollary, we could deduce that a degree n polynomial f has at most n roots.

1.3. Field extensions. We can now treat the first important definition. A *field extension* is a homomorphism of fields $i : k \rightarrow K$, which will often be denoted as K/k , suppressing the map i which is often clear from context. Note that we avoid working with inclusions as this is not a categorical notion. Furthermore, we will often want to think of k as being embedded in K in more than one way.

We remark that i makes K into a vector space over k and we will denote its dimension by $[K : k]$, which is called the *degree of the field extension*. If $[K : k] < \infty$ we shall say that K/k is a finite field extension.

Example 1.4. Some examples of field extensions are

$$\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C} \quad k \subseteq k(t) \quad \mathbb{Q} \subseteq \mathbb{Q}(i).$$

The finite field extensions among these are $\mathbb{R} \subseteq \mathbb{C}$ and $\mathbb{Q} \subseteq \mathbb{Q}(i)$.

Whenever we define a new object in mathematics, we are morally obligated to discuss the morphisms. Given two field extensions $k \rightarrow K$ and $k \rightarrow L$ we say that a ring homomorphism $\varphi : K \rightarrow L$ is a *k-homomorphism* if the following diagram commutes:

$$\begin{array}{ccc} K & \xrightarrow{\varphi} & L \\ \uparrow & \nearrow & \\ k & & \end{array}$$

In particular, it follows from this definition that a *k-homomorphism* is *k-linear* as a map between vector spaces over k . We shall denote the collection of *k-homomorphisms* by $\text{Hom}_k(K, L)$ and the *k-automorphisms* of K by $\text{Aut}_k(K)$.

Suppose that we have a field extension K/k and $\alpha \in K$. Then we define the evaluation homomorphism (which is *k-linear*) by

$$\begin{aligned} \text{ev}_\alpha : k[x] &\longrightarrow K \\ f(x) &\longmapsto f(\alpha). \end{aligned}$$

Notice that the image of this map is $k[\alpha] \subseteq K$ and that the kernel must be a prime principal ideal of $k[x]$ since $k[x]/\ker(\text{ev}_\alpha) \hookrightarrow K$.

- Case 1: If $\ker(\text{ev}_\alpha) = 0$ then $k[\alpha] \cong k[x]$ and hence α isn't the zero of any nonzero polynomial over k . In this case, we say that α is *transcendental over k*. Examples of this are $t \in k(t)$ over k and $e, \pi \in \mathbb{R}$ over $\mathbb{Q}$.
- Case 2: If $\ker(\text{ev}_\alpha) = (m_\alpha)$ for a unique monic irreducible polynomial $m_\alpha(x)$, then we say that m_α is the *minimal polynomial of α over k* and we say that α is algebraic over k . Note that for any $f \in k[x]$ we have that $f(\alpha) = 0$ if and only if m_α divides f and that $k[\alpha] = k(\alpha)$ is a field.

Exercise 1.5. Prove that if $\alpha \in K$ is algebraic over k , then

$$[k(\alpha) : k] = \dim_k(k[x]/m_\alpha(x)) = \deg(m_\alpha(x)).$$

2. LECTURE 2

2.1. Finite, algebraic and finitely generated field extensions. In this lecture, we will explore the relationship between different finiteness notions for field extensions. We will start with some obvious remarks.

We begin by remarking that if $\alpha \in K$ is algebraic over k , then $k[x]/(m_\alpha(x))$ is a field extension of k . Conversely, given a monic irreducible polynomial $f(x) \in k[x]$, we see that

$k_f := k[x]/(f(x))$ is a field extension of k of degree $\deg(f)$ and f has the zero $\bar{x} \in k_f$. In particular, by choosing an irreducible factor, we deduce the following proposition.

Proposition 2.1 (Kronecker). *For all nonconstant $f \in k[x]$ there exists a finite field extension K/k for which f has a root in K .*

We will also need the following easy proposition.

Proposition 2.2 (Tower law). *If $L/K/k$ are field extensions, then*

$$[L : k] = [L : K][K : k].$$

Proof. Pick bases (α_i) of L over K and (β_j) of K over k . It is not hard to show that $(\alpha_i\beta_j)$ is a basis of L/k . $\square$

This already gives us some interesting consequences. A classical application is to constructability questions using a compass and an unmarked ruler.

Exercise 2.3. In this exercise, you will prove that given a cube in $\mathbb{R}^3$ cannot be doubled using a compass and ruler. Consider the corners of the unit cube in $\mathbb{R}^3$ and note that the coordinates of each of these points are rational and suppose for the sake of contradiction that this cube can be doubled.

- At each step, we will consider the field generated by the coefficients of all the thus far constructed points. Show that when adjoining a new point, we get a field extension of degree at most 2.
- Show that the degree of the field generated by the coefficients of the thus constructed points over $\mathbb{Q}$ is always a power of 2.
- Argue that $x^3 - 2$ is the minimal polynomial of $\sqrt[3]{2}$ using Eisenstein's criteria and Gauss' lemma.
- Deduce, using the tower law, that $\sqrt[3]{2}$ is not in the field generated by the coefficients of all thus constructed points and conclude that the cube cannot be doubled.

Let us now introduce a couple of more definitions for field extensions K/k .

- If $K = k(\alpha_1, \dots, \alpha_n)$ for some $\alpha_i \in K$ we say that K/k is *finitely generated*.
- If $K = k(\alpha)$ for some $\alpha \in K$ we say that K/k is *simple* or *primitive*.
- If every element of K is algebraic over k we say that K/k is *algebraic*.

The proof of the following useful universal property follows immediately from the universal property of quotients.

Proposition 2.4. *Given a field extension K/k , a polynomial $f(x) \in k[x]$ and any $\alpha \in K$ such that $f(\alpha) = 0$ there exists a unique k -homomorphism*

$$\begin{aligned} \varphi : k_f &\longrightarrow K \\ \bar{x} &\longmapsto \alpha. \end{aligned}$$

We shall see later that the iterative use of this proposition can be used to construct elements of $\text{Gal}(K/k)$.

Having introduced two different but related notions, that of K/k being finite or finitely generated, we ought to compare them. For instance $k(t)$ is finitely generated over k , yet not finite. In general, we have the following.

Proposition 2.5. *The field extension K/k is finite if and only if it is finitely generated and algebraic.*

Proof. Suppose first that K/k is finite. Obviously, K/k is finitely generated. Furthermore, for any $\alpha \in K$, $1, \alpha, \dots, \alpha^d$ must be linearly dependent, where $d = [K : k]$, thus α is algebraic. This proves the forward direction.

Assume now that K/k is finitely generated and algebraic. Thus, there exist $\alpha_1, \dots, \alpha_n \in K$ such that $K = k(\alpha_1, \dots, \alpha_n)$. We will prove inductively that K is finite. Recall that we showed in the last lecture that primitive algebraic extensions are finite. This means that each extension $k(\alpha_1, \dots, \alpha_k)(\alpha_{k+1})/k(\alpha_1, \dots, \alpha_k)$ is finite (since algebraic over k implies algebraic over $k(\alpha_1, \dots, \alpha_k)$) and by the tower law, K/k is finite, as required. $\square$

Exercise 2.6. By examining the proof of Proposition 2.5, show that if $\alpha_1, \dots, \alpha_n \in K$ are algebraic over k , then $k(\alpha_1, \dots, \alpha_n)/k$ is an algebraic extension.

Exercise 2.7. If $\alpha_1, \dots, \alpha_n \in K$ are algebraic over k , then $k[\alpha_1, \dots, \alpha_n] = k(\alpha_1, \dots, \alpha_n)$.

We will finish this lecture with a last proposition.

Proposition 2.8. *If $L/K/k$ is a tower of field extensions, then*

- (1) *L/k is finite if and only if L/K and K/k are finite.*
- (2) *L/k is algebraic if and only if L/K and K/k are algebraic.*

Proof. Let $L/K/k$ be a tower of field extensions.

- (1) This follows from the tower law.
- (2) The forward direction is trivial. Assume that L/K and K/k are algebraic and let $\alpha \in L$ be arbitrary. Since L/K is algebraic, there exist $c_{n-1}, \dots, c_0$ not all equal to zero such that

$$\alpha^n + c_{n-1}\alpha^{n-1} + \dots + c_0 = 0.$$

The trick now, is to consider the field $K' = K(c_{n-1}, \dots, c_0)$. Since K'/k is finitely generated and algebraic, it is a finite extension by Proposition 2.5. Furthermore, $K'(\alpha)/K'$ is finite. By the tower law, $K'(\alpha)/k$ is finite and thus α is algebraic over k . It follows that L/k is algebraic, as required. $\square$

The slogan in the above proof is, as is often the case in algebra, to throw away all but finitely much data.

3. LECTURE 3

3.1. Lemmas. Let us first recall some easy facts.

Exercise 3.1. Let K/k and L/k be field extensions and $\varphi : K \rightarrow L$ be a k -homomorphism. Suppose that $f(x) \in k[x]$ and $\alpha \in K$ is a root of f . Then $\varphi(\alpha) \in L$ is also a root of f .

Lemma 3.2. *If K/k is algebraic and $\varphi : K \rightarrow K$ is a k -homomorphism, then φ is an automorphism.*

Proof. The slogan of this proof is to zoom in on only finitely much data. As a map of fields we already know that φ is injective. Let $\alpha \in K$ be arbitrary. Then, by the previous exercise,

$$X := \{\beta \in K : m_\alpha(\beta) = 0\}$$

is preserved by φ . Since X is finite and φ is injective, we have that φ is a permutation of X . It follows that α is in the image of φ and since α was arbitrary, φ is an automorphism. $\square$

The above result obviously fails if K/k is not algebraic. Consider for example the non-surjective $\mathbb{Q}$ -homomorphism

$$\begin{aligned}\varphi : \mathbb{Q}(t) &\longrightarrow \mathbb{Q}(t) \\ t &\longmapsto t^2.\end{aligned}$$

Exercise 3.3. If K/k is a field extension, then

$$L := \{\alpha \in K : \alpha \text{ is algebraic over } k\}.$$

We remark that this gives in particular that

$$\bar{\mathbb{Q}} := \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$$

is a field. In particular, it is an example of an *algebraic closure*.

3.2. Algebraic closure. The goal in this section is to show that, for any field k , there exists a maximal algebraic extension $\bar{k}/k$ and that this is unique up to k -isomorphism. However, it will not be unique up to unique isomorphism, meaning that there is no universal property for $\bar{k}$. This nonuniqueness is measured by the automorphism group $\text{Gal}(\bar{k}/k)$ and is what makes Galois theory interesting.

We see that if $\bar{k}$ exists, then the only irreducible polynomials in $\bar{k}[x]$ are linear.

A field K is called *algebraically closed* if it has any of the following equivalent properties:

- (i) Every algebraic extension of K is trivial.
- (ii) Every irreducible polynomial of $K[x]$ has degree 1.
- (iii) Every non-constant polynomial in $K[x]$ has a root in K .
- (iv) Every polynomial in $K[x]$ factors linearly over K .

As examples, we see that $\bar{\mathbb{Q}}$ and $\mathbb{C}$ are algebraically closed, while $\mathbb{R}$ and $\mathbb{F}_p$ are not.

An *algebraic closure of k* is an algebraic field extension $\bar{k}/k$ such that $\bar{k}$ is algebraically closed.

Theorem 3.4. Any field k has an algebraic closure $\bar{k}/k$.

Proof. Consider the massive ring

$$A := k[T_{f,i} : f \text{ is monic irreducible and } 1 \leq i \leq \deg(f)].$$

This is an attempt to add in all the missing zeroes. To get a field, we should quotient by a maximal ideal and to get $T_{f,i}$ to be zeroes of f we should quotient by an ideal containing the ideal I which is generated by all coefficients of polynomials

$$f(X) - \prod_i (X - T_{f,i})$$

To find a maximal ideal, all we need to do is verify that I is a proper ideal. Suppose not, then there exist finitely many $a_j \in A$ and c_j coefficients of polynomials $f_j(X) - \prod_i (X - T_{f_j,i})$ such that

$$1 = \sum_j a_j c_j.$$

Using Kronecker's theorem, we can by passing to a finite field extension K/k find zeroes $\alpha_{j,i} \in K$ such that

$$f_j(X) = \prod_i (X - \alpha_{j,i}).$$

If we order the zeroes of f_j consistently, we get a k -homomorphism

$$\begin{aligned}\varphi : A[X] &\longrightarrow K[X] \\ T_{f_j,i} &\longmapsto \alpha_{j,i}.\end{aligned}$$

Applying this to $1 = \sum_j a_j c_j$ we obtain that $1 = 0$ which cannot hold in a field K . Consequently, I is a proper ideal. By Zorn's lemma, there exists a maximal ideal $I \subseteq \mathfrak{m}$.

We can now choose

$$\bar{k} := A/\mathfrak{m}.$$

We claim that this is an algebraic extension. This follows from the fact that the generators of A are algebraic by construction and any extension with algebraic generators is algebraic.

To show that $\bar{k}$ is algebraically closed, note that any polynomial $f(x) \in k[x]$ splits linearly over $\bar{k}$ by construction. For a general $g(x) \in \bar{k}[x]$ we can find a zero $\alpha \in K$ in some finite extension $K/\bar{k}$ and then $g(x) \mid m_{\alpha,k}$. Since $m_{\alpha,k}(x)$ splits over K , then so does $g(x)$. $\square$

Exercise 3.5. Prove that if $k[x_i : i \in I]/k$ is a field extension and if each x_i is algebraic, then $k[x_i : i \in I]/k$ is an algebraic extension. *Hint:* Use the corresponding fact when I is finite.

4. LECTURE 4

4.1. Algebraic closures. In this lecture, we will prove the uniqueness up to isomorphism of the algebraic closure of a field k .

Proposition 4.1. *Let K/k and L/k be field extensions.*

- (i) *If K/k is algebraic and L/k is algebraically closed, then there exists a k -morphisms $K \rightarrow L$.*
- (ii) *If K/k and L/k are algebraic closures, then there exists a k -isomorphism $K \cong L$.*

Proof. Suppose first that K/k is algebraic and that L/K is algebraically closed. Consider the set

$$X := \{(F, \theta) : k \hookrightarrow F \xrightarrow{\theta} K\}.$$

This is clearly nonempty and we can give a partial order by $(F_1, \theta_1) \leq (F_2, \theta_2)$ whenever $F_1 \subseteq F_2$ and $\theta_1 = \theta_2|_{F_1}$. Notice that any chain has an upper bound, since if $(F_i, \theta_i)_{i \in I}$ has an upper bound $(F, \theta) \in X$ with

$$F = \cup_i F_i, \quad \theta(x) = \theta_i(x) \text{ for } x \in F_i.$$

By Zorn's lemma, there exists a maximal element $(F, \theta) \in X$.

We claim that $F = K$, which would finish the proof. If not, then there exists $\alpha \in K \setminus F$. Since α is algebraic over k it satisfies a minimal polynomial m_α and this polynomial has a zero $\beta \in L$ because L is algebraically closed. Hence, by an earlier universal property there is a k -homomorphism $F(\alpha) \rightarrow L$, which is a contradiction. This proves (i).

In particular, if K/k and L/k are algebraic closures then there is a k -homomorphism $K \rightarrow L$. Since L/k is algebraic, L/K is algebraic and thus this gives us that $K \cong L$ as required, because K is algebraically closed. $\square$

We remark the following:

- If K/k is such that K is algebraically closed, then $L := \{\alpha \in K : \alpha \text{ algebraic over } k\}$ is the algebraic closure of k .
- If $k \subseteq K \subseteq \bar{k}$, then $\bar{K} = \bar{k}$.
- If $k \subseteq K \subseteq \bar{K}$ and K/k is algebraic, then $\bar{k} = \bar{K}$.

4.2. Splitting fields. Let K/k be a field extension. If $f(x) \in k[x]$ can be factored linearly in $K[x]$, we say that f *splits* over K . In case f splits over K and $K = k(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n \in K$ are the zeroes of f , we shall call K/k a splitting field of f .

We remark that:

- K/k is a splitting field if and only if it is minimal in the sense that $k \subseteq L \subseteq K$ and f splits over L implies that $L = K$.
- If f splits over L/k , then L contains a unique splitting field of f .
- Splitting fields are finite.
- If f splits over K and $g \mid f$ in $k[x]$, then g also splits over K .

Proposition 4.2. *Let f be a nonconstant polynomial. Then, f admits a splitting field, which is unique up to isomorphism.*

Proof. Since f splits over $\bar{k}$, we see that f has a splitting field inside of $\bar{k}$, which proves existence. Suppose that K_1/k and K_2/k are splitting fields and consider a k -isomorphism $\phi : K_1 \rightarrow K_2$. It is obvious that $\phi(K_1) \subseteq K_2$. Similarly $\phi^{-1}(K_2) \subseteq K_1$, implying $K_2 = \phi(K_1)$. Thus, $\phi|_{K_1}$ is an isomorphism between the splitting fields. $\square$

Example 4.3. A splitting field of $x^3 - 2$ is $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, where $\zeta_3 = e^{2\pi i/3}$. We have the tower

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \zeta_3).$$

Since the first extension has degree 3 and the second extension has degree 2, it follows from the tower law that $[\mathbb{Q}(\sqrt[3]{2}, \zeta_3) : \mathbb{Q}]$.

Exercise 4.4. Let f be a polynomial of degree n . Prove that a splitting field has degree dividing $n!$.

An algebraic extension K/k will be called *normal* if any irreducible polynomial $f(x) \in k[x]$ with a root in K splits over K .

Theorem 4.5. *Let K/k be a finite extension. Then, the following are equivalent:*

- (i) K/k is normal,
- (ii) K is a splitting field of some $f(x) \in k[x]$,
- (iii) any k -homomorphism $\phi : K \rightarrow \bar{K}$ satisfies that $\phi(K) = K$.

Proof. Suppose that the finite extension K/k is normal and is thus of the form $K = k(\alpha_1, \dots, \alpha_n)$. We see that K is a splitting field of $m_{\alpha_1}(x) \cdots m_{\alpha_n}(x)$. This shows that (i) $\implies$ (ii).

Suppose that K is a splitting field of a nonconstant $f(x) \in k[x]$ and consider a k -homomorphism $\phi : K \rightarrow \bar{K}$. Then, $\phi(K) = K$, since $K[x]$ is a UFD. This proves (ii) $\implies$ (iii).

Suppose that any k -homomorphism $\phi : K \rightarrow \bar{K}$ satisfies that $\phi(K) = K$. Suppose that an irreducible $f(x) \in k[x]$ has a root $\alpha \in K$ and let $\beta \in \bar{K}$ be any other root. Then, we have a map

$$\begin{aligned} k(\alpha) &\longrightarrow \bar{K} \\ \alpha &\longmapsto \beta \end{aligned}$$

by a universal property. By an earlier extension proposition, we deduce that there is a map $\theta : K \rightarrow \bar{K}$ such that $\theta(\alpha) = \beta \in K$, which finishes the proof of (iii) $\implies$ (i). $\square$